

H&M Dataset

**EDA highlights and modeling
implications**

Part I

Articles

Why this table matters

This is the **item catalog** for the H&M recommendations problem:

- **SKU-level** rows (`article_id`) with rich categorical metadata
- Text field (**`detail_desc`**) for semantic signals
- Multiple “code + name” pairs that must stay consistent

Articles at a glance

Metric	Value
Rows	105,542
Columns	25
Dtypes	11 int64, 14 object
Duplicate article_id	0
Missing detail_desc	416 (0.39%)

All in all: dense catalog, very clean, plus a tiny text-missing pocket.

Schema (a bit hard)

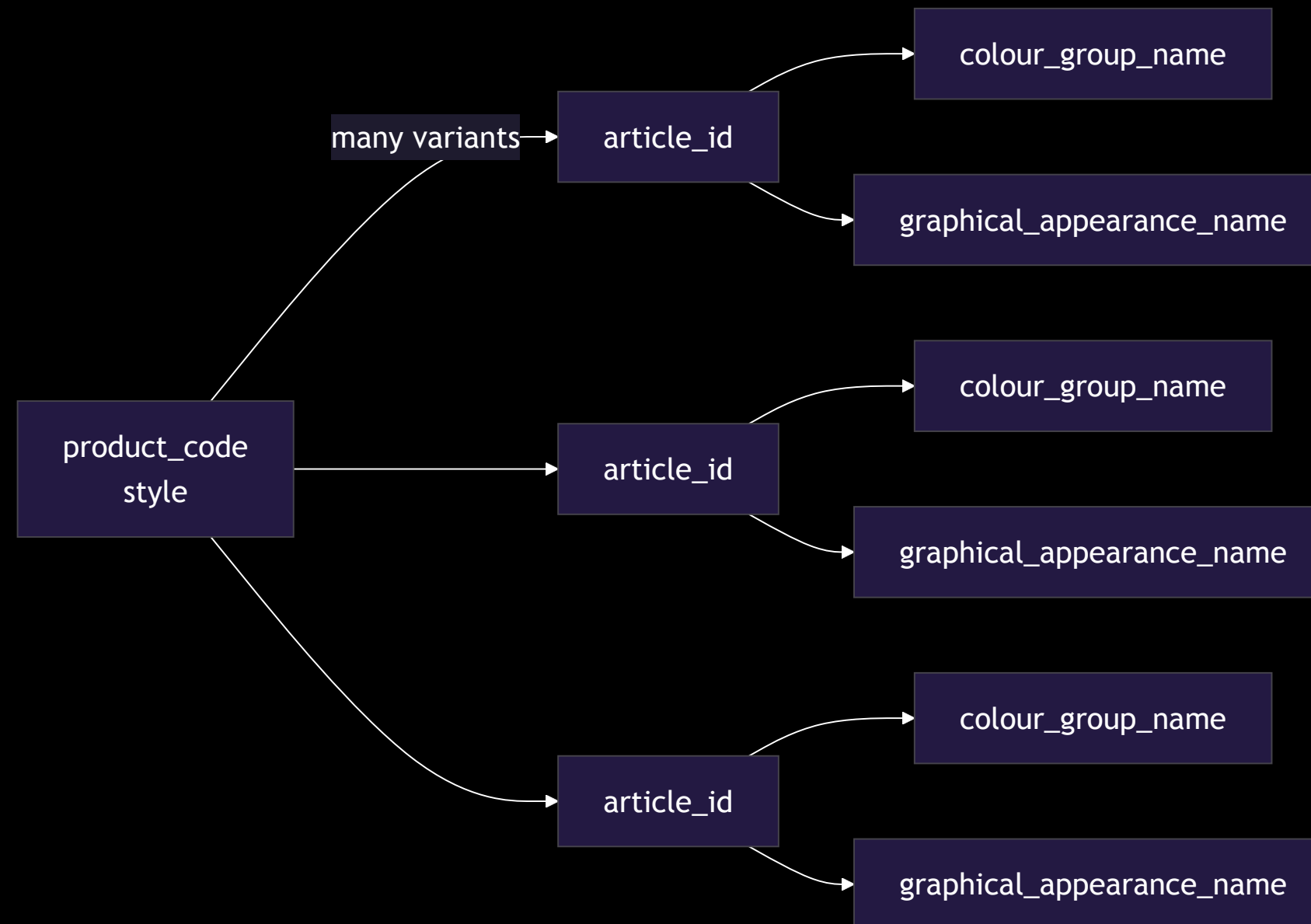
Identifiers

- article_id (SKU identifier)
- product_code (style identifier grouping multiple variants)

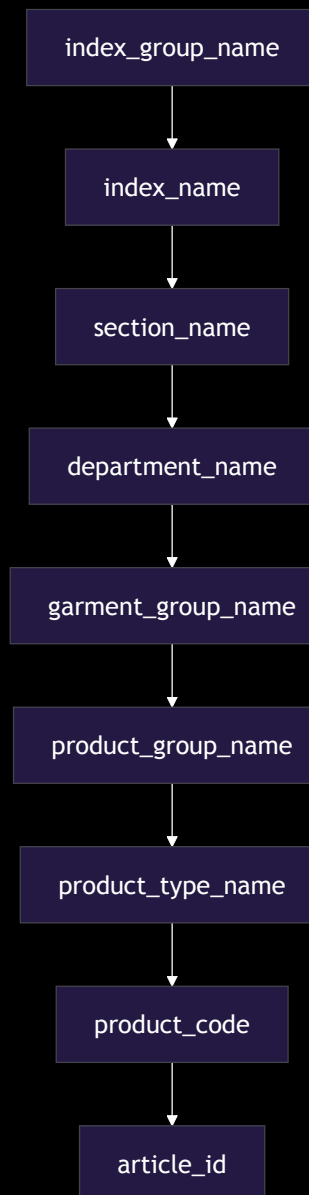
Naming & taxonomy

- Product: prod_name, product_type_*, product_group_name
- Visuals: graphical_appearance_*, colour_group_*, perceived_colour_*
- Merch hierarchy: department_*, section_*, index_*, index_group_*, garment_group_*
- Text: detail_desc

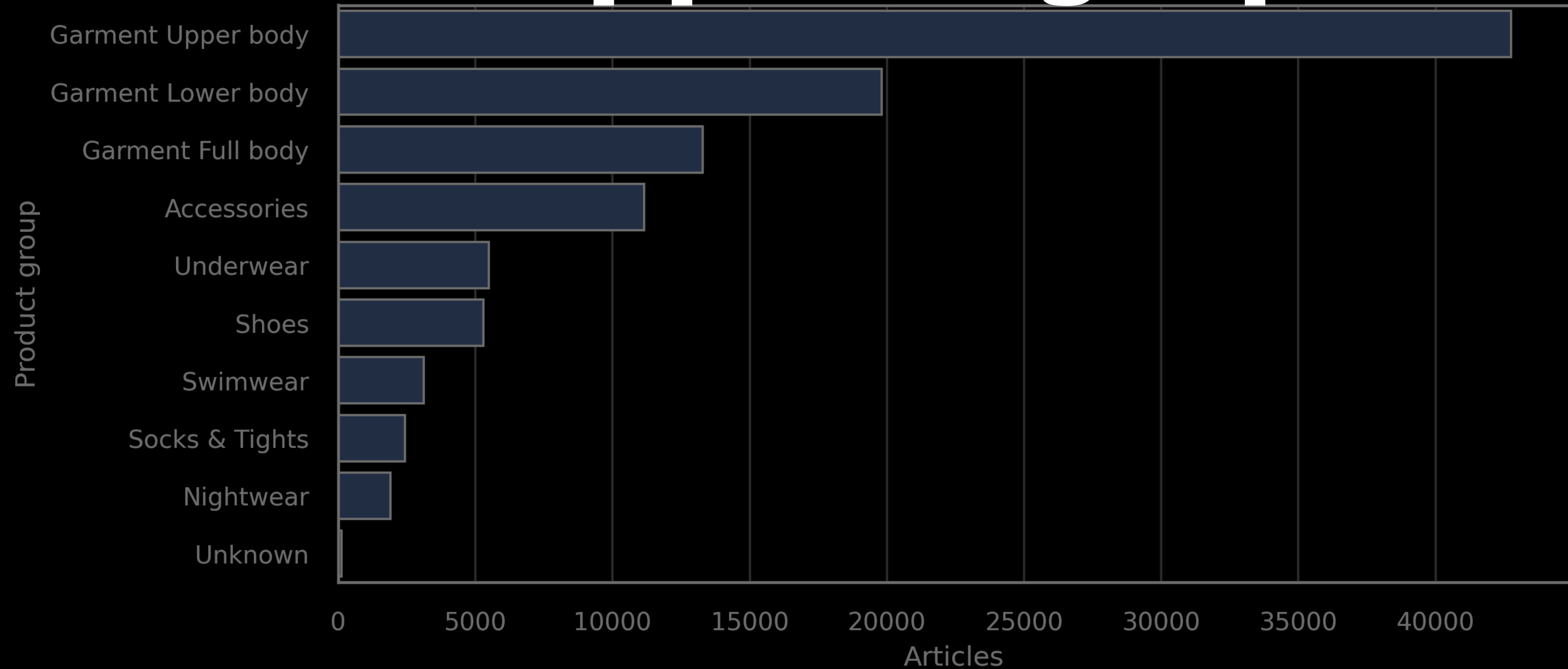
Variant structure (style → SKUs)



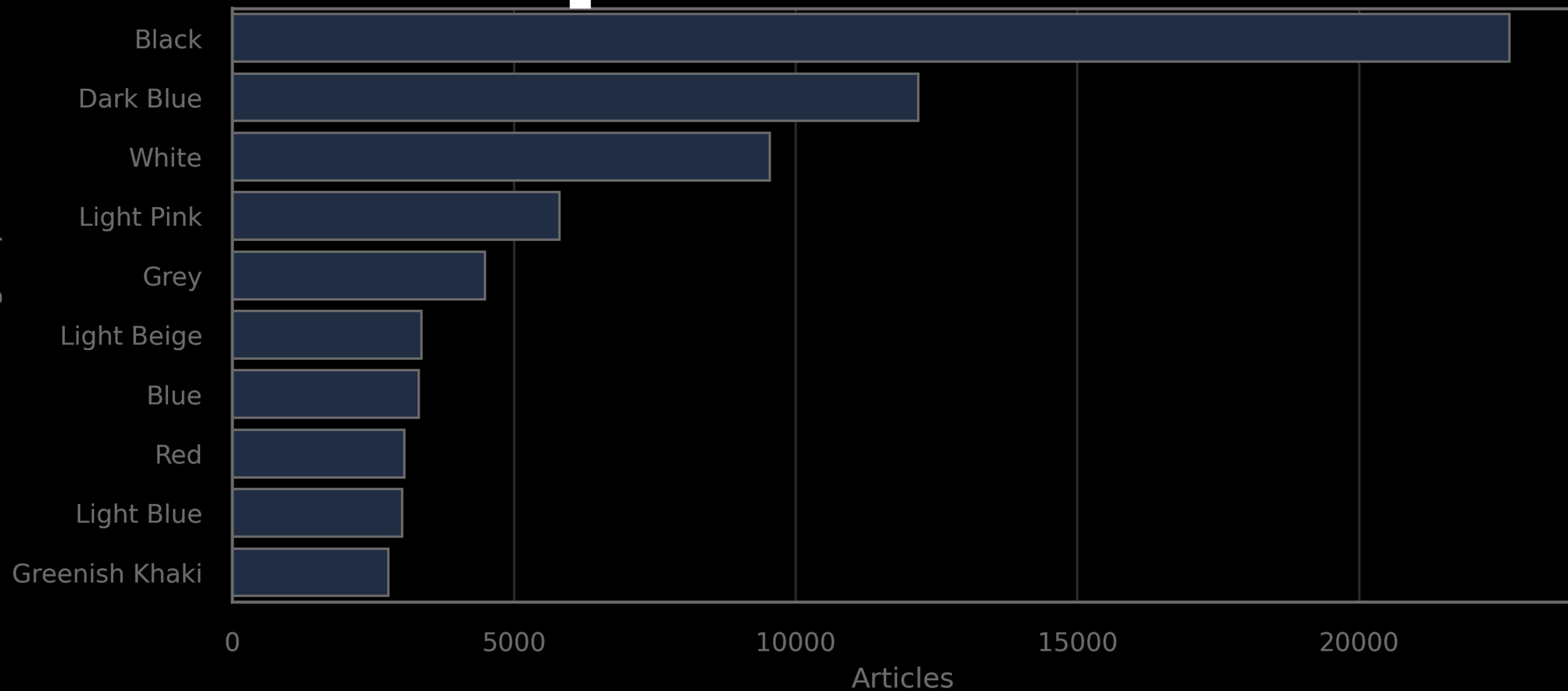
Merch taxonomy stack (where labels come from)



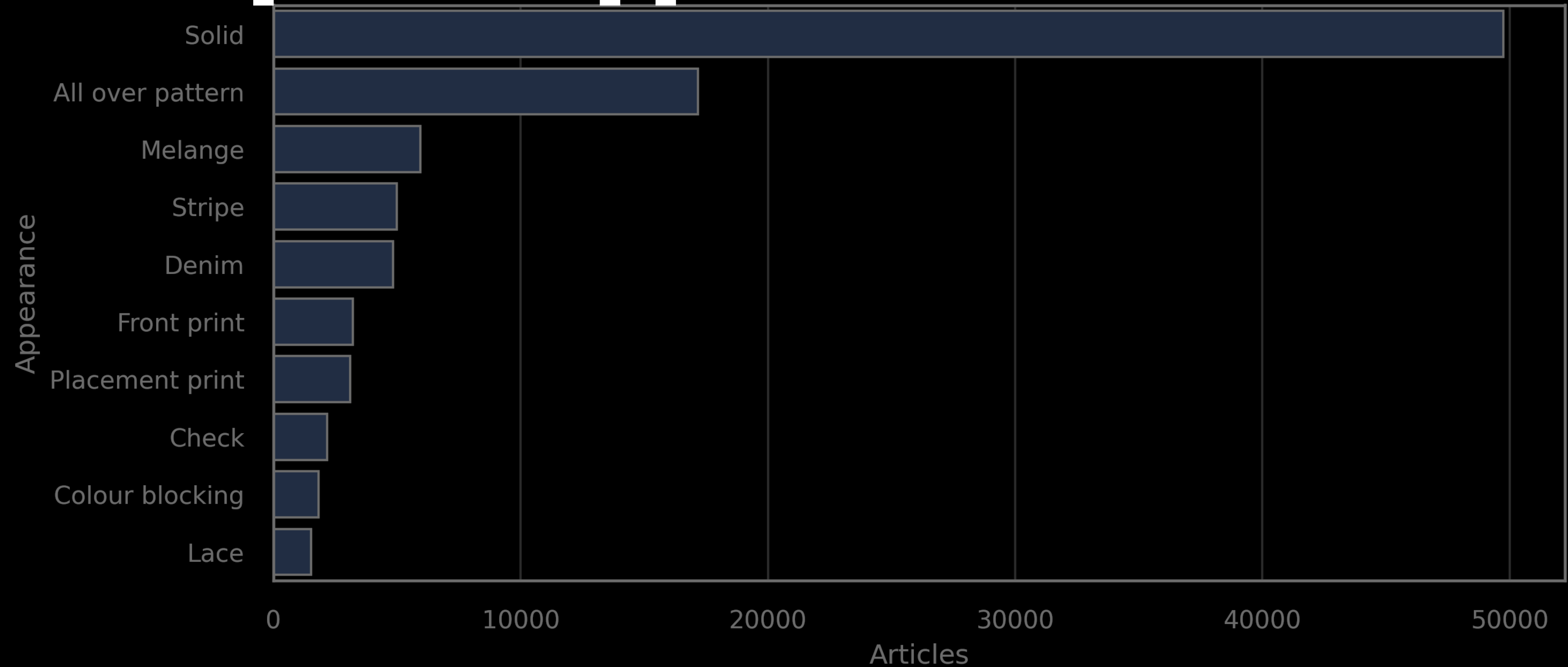
Articles: top product groups



Articles: top colors



Graphical appearances



Part II

Customers (profiles and signals)

Customers dataset snapshot

Metric	Value
Rows	1,371,980
Columns	7
Duplicate customer_id	0

Columns:

customer_id, FN, Active, club_member_status,
fashion_news_frequency, age, postal_code

Columns

- customer_id: anonymized unique user key (join key to transactions)
- FN: **semi-observed flag** (1.0 or NaN)
- Active: **semi-observed flag** (1.0 or NaN)
- club_member_status: membership lifecycle state (ACTIVE, PRE-CREATE, LEFT CLUB, missing)
- fashion_news_frequency: newsletter frequency preference (NONE, Regularly, Monthly, missing)
- age: numeric demographic feature (mostly present)
- postal_code: anonymized location proxy, very high-cardinality

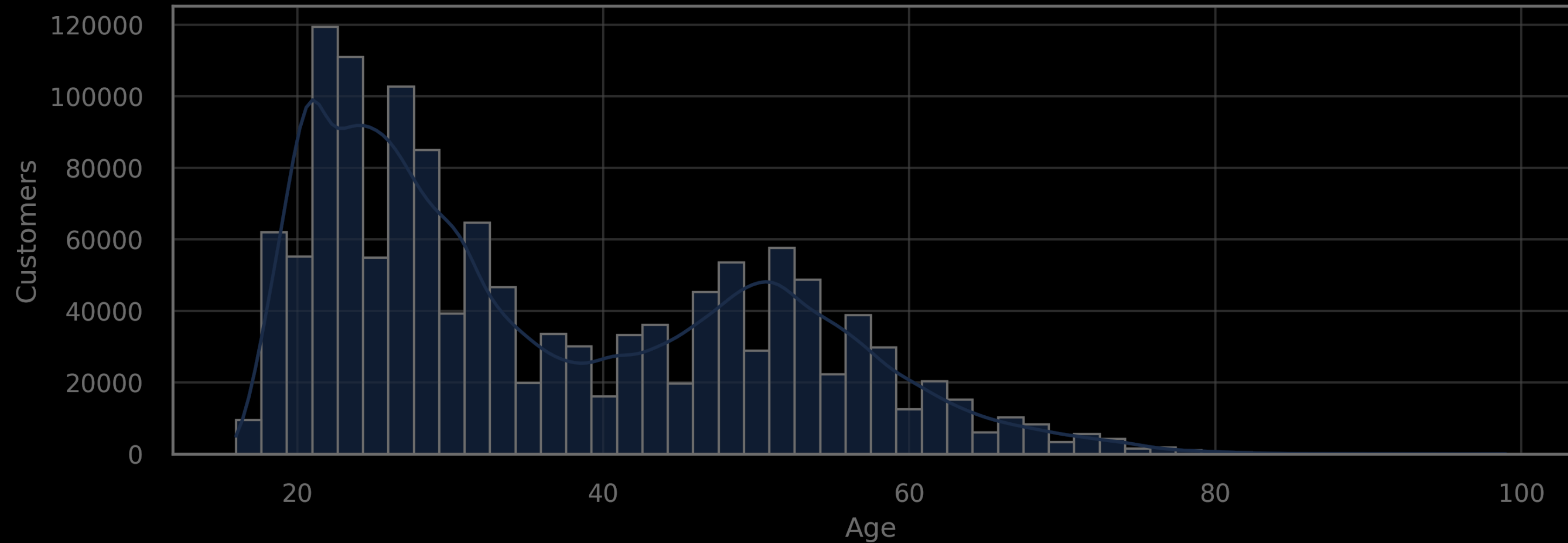
Missingness is the story here

Non-zero missingness:

- **FN missing: 65.24%**
- **Active missing: 66.15%**
- fashion_news_frequency missing: 1.17%
- age missing: 1.16%
- club_member_status missing: 0.44%
- FN/Active are not “broken data”, they encode a sparse flag system.

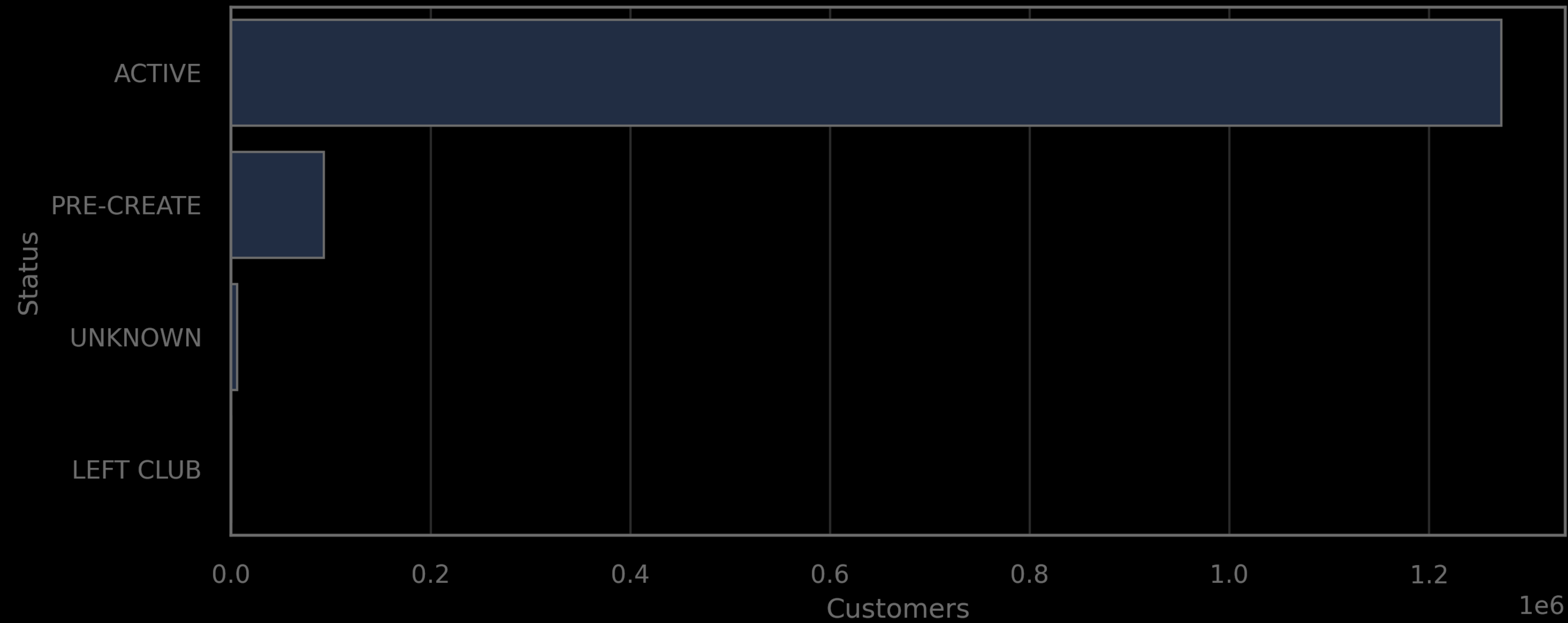
Customers: age distribution

Customers: age distribution



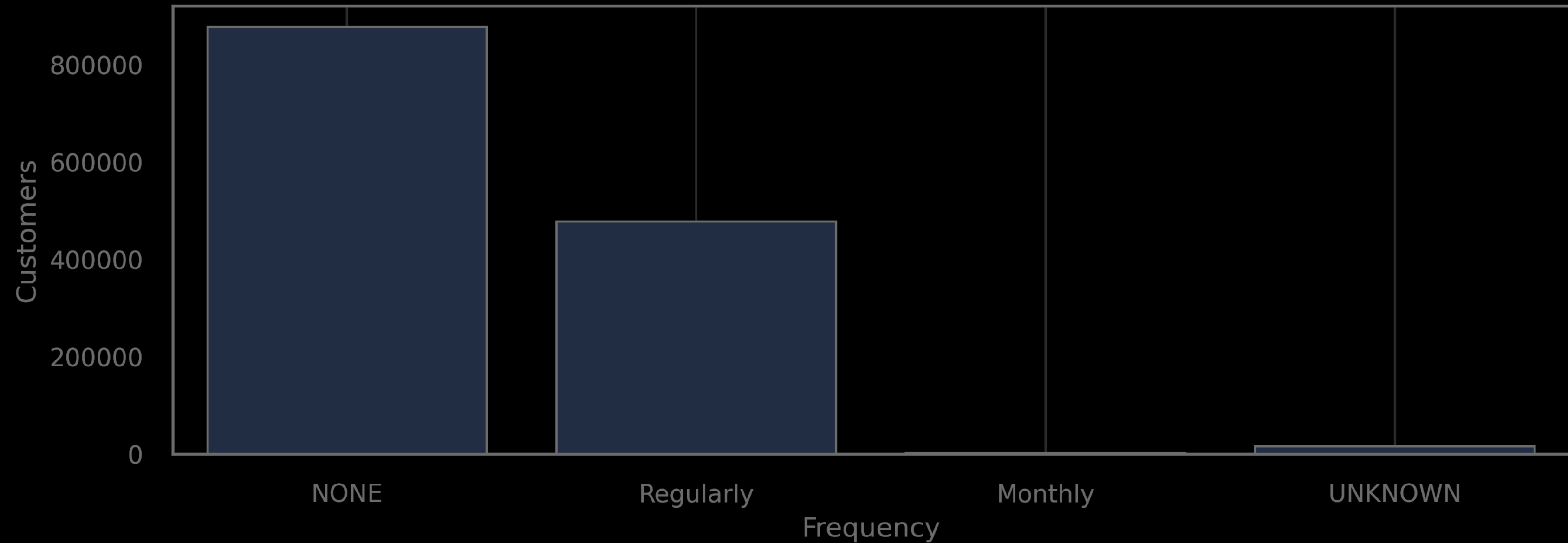
Customers: club status

Customers: club_member_status

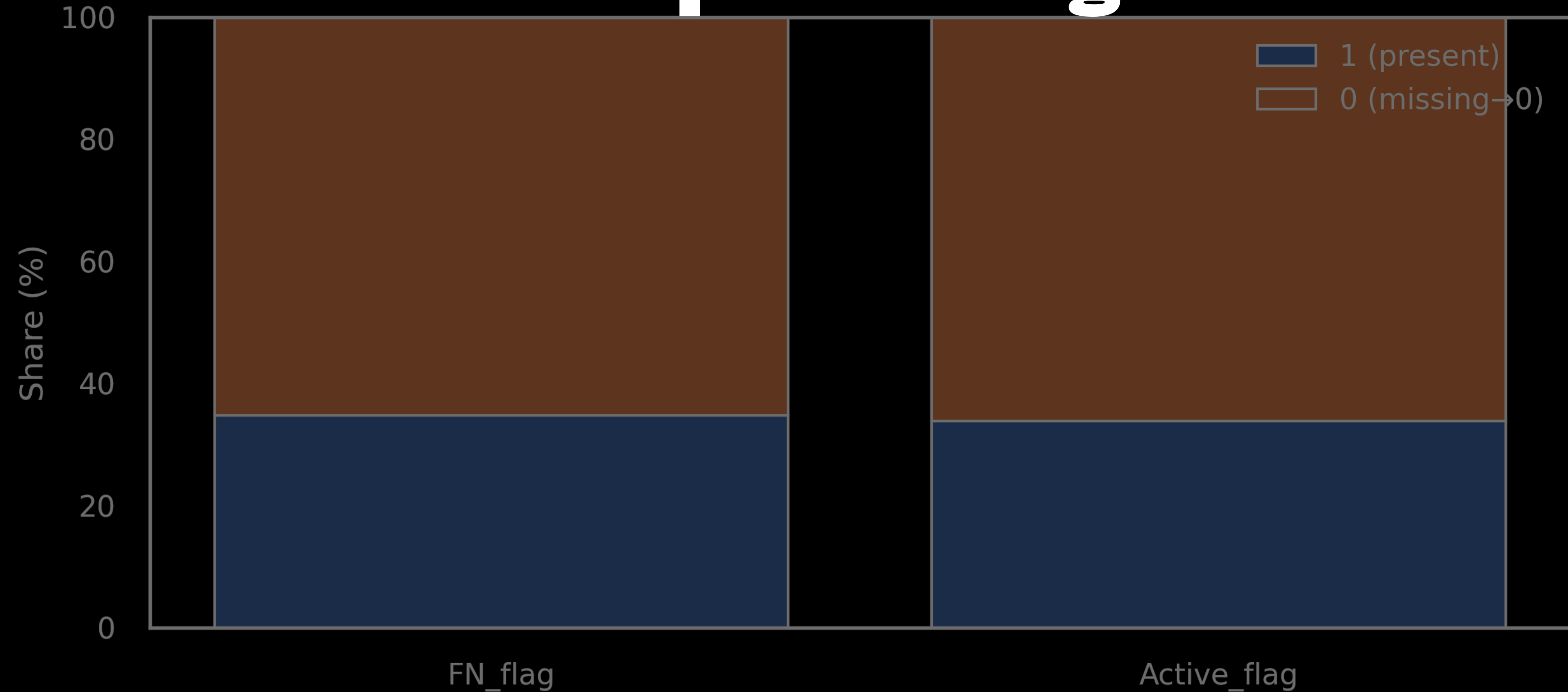


Customers: news frequency

Customers: fashion_news_frequency



Customers: sparse flags



FN and Active are not numeric variables

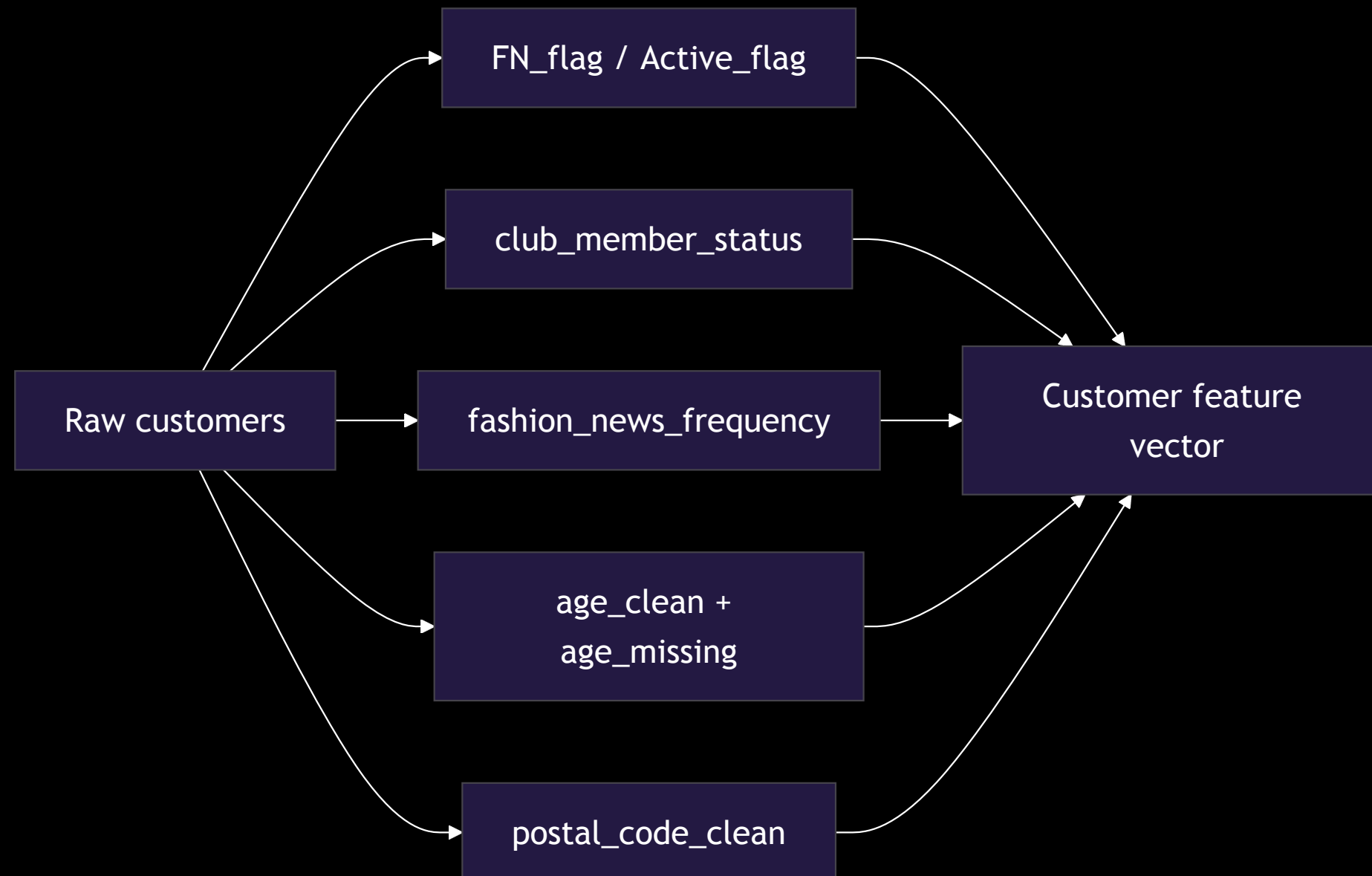
Cardinality check:

- FN nunique: **1**
- Active nunique: **1**
- when present, the value is always 1.0
- everything else is NaN

So we treat them as **binary flags**.

```
df["FN_flag"] = (df["FN"] == 1).fillna(False).astype(int)
df["Active_flag"] = (df["Active"] == 1).fillna(False).astype(int)
```

Customers: feature blueprint



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